# Diagonal concavity does not imply global concavity

A counterexample to MathDB 338076

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**Abstract**

We disprove Lachaume's diagonal-concavity conjecture with four variables and strictly positive integer coefficients. The fourth root is strictly concave on the entire positive diagonal, but has a strictly positive second directional derivative at a point of the positive orthant. The proof consists of two explicit polynomial identities.

## Task 1: full resolution

For $x = (x_1, \dots, x_n)$, let

$$\sigma_k(x) = \sum_{1 \le i_1 < \cdots < i_k \le n} x_{i_1} \cdots x_{i_k}, \qquad \sigma_0(x) = 1.$$

Conjecture 1 of [1], stated on pages 3 and 8 and repeated on page 18, asserts that for nonnegative coefficients $a_0, \dots, a_p$, concavity of

$$\left( \sum_{k=0}^{p} a_k \sigma_k(x) \right)^{1/p}$$

on the diagonal $x_1 = \cdots = x_n > 0$ is equivalent to concavity on $\Gamma_n = (0, \infty)^n$. Here $1/p$-concavity means concavity of the $1/p$ power, as in the source's Definition 1.2.

**Theorem 1.** *Take $n = p = 4$ and*

$$F(x) = 20 + 40\sigma_1(x) + 21\sigma_2(x) + 40\sigma_3(x) + 20\sigma_4(x). \tag{1}$$

*Then $t \mapsto F(t,t,t,t)^{1/4}$ has negative second derivative for every $t > 0$, whereas $F^{1/4}$ is not concave on $(0, \infty)^4$. Consequently $\mathcal{K}_4^4 \neq \mathcal{X}_4^4$, and the conjecture is false.*

*Proof.* For any positive twice differentiable function $h$ of one real variable,

$$(h^{1/4})'' = \frac{4hh'' - 3(h')^2}{16h^{7/4}}. \tag{2}$$

On the positive diagonal, the binomial counts in $\sigma_k$ give

$$Q(t) := F(t,t,t,t) = 20t^4 + 160t^3 + 126t^2 + 160t + 20.$$

Direct differentiation and expansion give the identity

$$3(Q')^2 - 4QQ'' = 48\big[1180(t^2-1)^2 + 80t(t^2+1) + 83t^2\big]. \tag{3}$$

Its right-hand side is strictly positive whenever $t > 0$. Since $Q(t) > 0$, formula (2) proves strict concavity on the whole positive diagonal.

Next set

$$x_0 = (2, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}), \qquad v = (1, 1, 1, 1), \qquad H(u) = F(x_0 + uv).$$

This line lies in the positive orthant for $u > -1/2$. Substitution into (1) yields

$$H(u) = 20u^4 + 230u^3 + 621u^2 + 713u + \frac{1235}{4}. \tag{4}$$

In particular,

$$H(0) = \frac{1235}{4}, \qquad H'(0) = 713, \qquad H''(0) = 1242,$$

and hence

$$4H(0)H''(0) - 3H'(0)^2 = 1235 \cdot 1242 - 3 \cdot 713^2 = 8763 > 0. \tag{5}$$

Formula (2) now gives

$$\frac{d^2}{du^2} F(x_0 + uv)^{1/4}\bigg|_{u=0} = \frac{8763}{16(1235/4)^{7/4}} > 0.$$

The restriction of a concave function to any line segment in its domain is concave, and a twice differentiable concave function has nonpositive second derivative. This contradiction proves the assertion. □

**Scope.** Every coefficient and every coordinate in the example is strictly positive. The failure occurs inside the domain, while diagonal concavity is strict everywhere. No endpoint convention or zero coefficient is used. One counterexample settles the universally quantified equivalence; the forward restriction implication remains valid.

The weaker real-rootedness criterion is a distinct question. Indeed, for $t \neq 0$ we have

$$\frac{Q(t)}{t^2} = 20w^2 + 160w + 86, \qquad w = t + t^{-1}.$$

The quadratic in $w$ has a root in $(-2, 2)$, since its values at $-2$ and $2$ are $-154$ and $486$. The corresponding roots of $t^2 - wt + 1$ are nonreal. Thus $Q$ is not real-rooted, and the example does not challenge the real-rootedness sufficient condition in [2, Theorem 2.7(d)]. It is also distinct from the derivative convolution result previously recorded under MathDB 338079.

## Proposed follow-up tasks

The original conjecture has been fully refuted above. The following are four further research tasks, not additional solution claims.

2. Determine whether diagonal 1/3-concavity implies global 1/3-concavity for every $n \geq 3$ and every strictly positive coefficient vector $(a_0, a_1, a_2, a_3)$.

3. Give necessary and sufficient conditions on positive $(a_0, a_1, a_2, a_3, a_4)$ for $(\sum_{k=0}^{4} a_k \sigma_k)^{1/4}$ to be concave on $(0, \infty)^4$.

4. Determine all $b > 0$ for which $(1 + 2\sigma_1 + b\sigma_2 + 2\sigma_3 + \sigma_4)^{1/4}$ is concave on $(0, \infty)^4$; compare this set with the parameters giving concavity on the positive diagonal.

5. Decide whether for every pair of integers $n \geq p \geq 4$ there are strictly positive coefficients for which $(\sum_{k=0}^{p} a_k \sigma_k)^{1/p}$ has negative second derivative everywhere on the positive diagonal but is not globally concave.

**Verification and attribution.** The accompanying standard-library Python verifier reconstructs both restrictions from elementary-symmetric sums and checks (3)–(5) in exact rational arithmetic. The argument is a complete analytic counterexample, not an inference from sampled values. This manuscript received author-agent self-review; independent peer review and proof-assistant verification are not claimed. The dated literature audit states the public sources checked and its access limits.

## References


[1] X. Lachaume, *On the concavity of a sum of elementary symmetric polynomials*, arXiv:1712.10327v1 (2017), `https://arxiv.org/abs/1712.10327`.

[2] X. Lachaume, *The constraint equations of Lovelock gravity theories: a new $\sigma_k$-Yamabe problem*, arXiv:1712.04528v2 (2018), `https://arxiv.org/abs/1712.04528`.